Sam McClenaghan

780-221-1327 | sam@aream.ca | smccl.ca

Experience

Software Engineer, Intern May 2026 – Present
Vertical City *Victoria, BC*

- Built an AWS media pipeline on S3 and CloudFront serving 2M+ requests/day across 10,000+ screens, replacing a paid vendor and saving $20K/year
- Cut media processing latency from 5 minutes to seconds with event-driven, in-process rendering
- Cut bandwidth 20% with ffmpeg perceptual hashing, catching re-encoded duplicates checksums miss
- Built rule-based targeting that matches content feeds to displays by city, region, and language

Full-Stack Developer, Intern May 2025 – Dec 2025
Leanpub *Remote (Victoria, BC)*

- Migrated 100+ pages from a Rails 4 monolith to Rails 8, GraphQL, and React Router 7, shipping both the backend and the front end for each
- Led a payment processing refactor unifying handling across product types, fixing scattered edge cases and enabling reliable refunds and purchase emails
- Built real-time job progress tracking on Redis and background workers with a React progress bar

Software Developer, Intern Nov 2022 – Aug 2024
Trust Science *Edmonton, AB*

- Designed a version-control system for client scoring configs, storing changes as JSON Patch deltas
- Built an Elasticsearch logging pipeline on DynamoDB streams to index logs from ML scoring jobs
- Migrated a 40+ page front end from Vue 2 and Bootstrap to Vue 3 and Tailwind CSS

Education

University of Victoria Victoria, BC
Honours Bachelor of Science *Sep 2023 – Present*
Computer Science major. Mathematics minor. GPA 7.8/9. Director of Projects at VikeLabs.

Projects

CourseFlow | *TypeScript, React, Cloudflare Workers, D1* | courseflow.smccl.ca 2025 – 2026

- Built a course planner used by 2,000+ UVic students, generating 3,000+ shareable timetables
- Kept shared timetables in sync live, pushing updates over WebSockets backed by Durable Objects
- Built course recommendations combining BM25 lexical search with vector embeddings

Anchor | *Next.js, TypeScript, Supabase, Anthropic SDK, Resend* 2025

- Built a mental-health companion at UVicHacks × Claude that detects distress from HRV data
- Integrated the Anthropic SDK to deliver Claude-powered grounding during distress episodes
- Owned backend and DevOps: Supabase/Postgres migrations, Vercel deployment, and Garmin ingestion

Technical Skills

Languages: Ruby, Go, TypeScript, JavaScript, Python, Java, C, SQL
Frameworks: Ruby on Rails, RSpec, React, Next.js, Remix, Vue, Tailwind CSS
Cloud & Tools: AWS (S3, Lambda, MediaConvert, CloudFront, DynamoDB), Docker, ffmpeg
Databases: MySQL, PostgreSQL, Redis, Elasticsearch